B-cell epitope prediction from antigen-only dynamics: project log

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Living record. Read top to bottom to get up to speed.

Running record of the project: predict B-cell epitope residues from *antigen-only* information, using alanine-scan $\Delta\Delta G_{\text{bind}}$ as residue-level binding-importance labels and asking whether antigen dynamics / solvation / energetics carry predictive signal *beyond static surface exposure*. Read in order; each section is one iteration — what we tried, what we found, what it changed. Not a formal manuscript. Every figure is regenerated by a named script (recorded in `figures/FIGURES.md`); every number traces to a file or command; unconfirmed values are flagged `[verify]`.

1 Motivation and the gap

We want to predict which residues of an antigen are B-cell epitope residues, from the antigen structure alone, in cases where the antibody is unknown. Doing this rigorously requires residue-level labels for "binding importance." We take these from **alanine scanning**: for residue i, $\Delta\Delta G^{(i)}_{\text{bind}} = \Delta G^{\text{mut},i}_{\text{bind}} - \Delta G^{\text{WT}}_{\text{bind}}$, with large positive values flagging residues whose wild-type identity stabilizes the interface. We treat this as a residue-level epitope score, $S_i^{\text{epitope}} \approx \Delta\Delta G^{(i)}_{\text{bind}}$. (Full derivation and the alchemical / StaB-ddG label tiers: `../../tcr/docs/b_cell_eptiope_prediction_may23_update_final.pdf`

The gap. Most B-cell epitope predictors — and the benchmarks that train them — label residues by *contact distance* to a bound antibody. As a coarse first pass this is defensible: proximity to the paratope is a reasonable way to *nominate* candidate interface residues. The limitation is treating every contact as a positive label of equal weight, which conflates "close" with "important": many interface residues contribute little binding free energy, while a few hotspots carry most of the affinity. The bias is baked into the data itself. Of the 4,164 conformational (discontinuous) B-cell epitopes in IEDB [1] that carry a deposited structure, essentially all (4,161; $> 99\%$) are defined by crystallographic or cryo-EM *contact* rather than by any binding measurement;[1] models trained on these labels therefore inherit proximity as their working definition of "epitope." Alanine-scan $\Delta\Delta G$ supplies the missing energetic refinement — *which* of the contacting residues actually pay for binding. Even a binary critical/non-critical call from a saturating alanine scan is energetically grounded, since "critical" means mutation measurably weakens binding; proximity and energetics are thus complementary, the former a candidate filter and the latter the importance label.

Separately, since Westhof et al. [2] it has been known that antigenicity correlates with *segmental mobility* (flexibility / B-factor) — an antigen-intrinsic, antibody-agnostic property — and

[1] IEDB next-generation query API (`query-api.iedb.org`), `bcell_search`, structure type *Discontinuous peptide* with a non-null `pdb_id`, queried 2026-06. IEDB attaches a structure only when the epitope was itself defined from one, so structure-linked records are overwhelmingly contact-derived; the mutation/binding-derived epitopes carry no `pdb_id`.

energy-decomposition analyses of plain MD trajectories can already localize antibody epitopes on an antigen without knowledge of the antibody [3]. The opportunity is to test, with modern MD, whether antigen-only *dynamical* and *solvation* features (RMSF, collective-mode participation, hydration thermodynamics) predict energetic binding importance — and whether they beat static exposure. This is where the author's methods background (FRESEAN collective-mode analysis; 3D-2PT-style hydration thermodynamics; non-equilibrium energy flow) is the differentiator (`../../_literature/my_manuscripts/`).

2 Design decisions to hold onto

Five issues decide whether the project produces a result or an artifact. Recorded here so they stay front-of-mind.

1. **The label is antibody-specific; the features are antibody-agnostic.** $\Delta\Delta G^{(i)}_{\text{bind}}$ is a residue's contribution to *one* paratope, but $\mathbf{x}_i$ comes from the apo antigen and cannot know the antibody. So $f(\mathbf{x}_i) \to \Delta\Delta G^{(i)}$ can only recover the *antibody-marginal* (antigen-intrinsic) hotspot propensity. This is not a weakness to hide — it is exactly why dynamics/flexibility features should help (cf. Westhof). Where an antigen has several antibodies, average $\Delta\Delta G^{(i)}$ over them to *define* the intrinsic target.

2. **"Beyond SASA" is the primary hypothesis, not a footnote.** Surface exposure is the master confounder: exposed residues are simultaneously more likely to be hotspots, more flexible, and more hydrated. The publishable claim is not "flexibility predicts epitopes" (Westhof has that) but "dynamics/solvation features add signal *after* controlling for static exposure." Design: baseline = SASA + depth + residue type; then test the partial R^2 added by RMSF, mode participation, and 3D-2PT hydration.

3. **Effective $N \approx$ #antigens/folds, not #residues.** Residues within an antigen are correlated and the intrinsic signal is per-fold. Split **leave-one-complex-out** (ideally leave-one-fold-out); never random residue splits. At this scale the work is hypothesis-generating — powered for a strong dynamics-vs-exposure effect, not a deployable predictor. State this plainly.

4. **Features come from the apo antigen** — which is the point and sets a ceiling. Cryptic / induced-fit epitopes (conformational selection vs. induced fit) cannot be predicted in principle from apo features. Conformational-selection epitopes *should* show in apo flexibility/mode features; pure induced-fit ones will not. That contrast is itself a result.

5. **The MD protocol is not one stack.** A flexibility/RMSF run differs from a FRESEAN/3D-2PT run (the latter needs high-frequency velocity output, constraint removal, and a water model validated for dynamics). Likely two protocols per antigen. The house TIP3P/2 fs/LINCS default is scrutinized accordingly (see `../docs/METHODS.md`).

3 The label benchmark

Curated experimental alanine-scan $\Delta\Delta G$ for antigen-side residues lives in `benchmark/antigen_alanine_scanning` (6 sheets: README, Simple_Primary, Full_Schema [51 cols], hGH_Table2_4, Paper_Inventory, Counts). Audited 2026-06-18.

Coverage. 203 primary rows, 13 datasets, 7 antigens. Source: AB-Bind [4] (195 rows; `github.com/sarahsirin/` and the Jin hGH alanine scan [5] (8 rows; benchmark cites the 1994 thesis, Table 3.2). The dataset was subsequently expanded with three antigen-side scans from SKEMPI 2.0 [6] and a binary label tier from IEDB (§5 and the Tier-2 section). Sign convention: $+\Delta\Delta G \Rightarrow$ weaker binding $\Rightarrow$ residue important (38 negative = binding-improving alanines; 17 zeros).

Integrity. The $\Delta\Delta G \leftrightarrow K_d$-ratio arithmetic is internally consistent for all 203 rows ($< 2\%$ vs $\exp(\Delta\Delta G/RT)$). No censoring in the primary set. The 4 non-alanine rows (hGH, EC50-derived) are correctly flagged `exclude_from_binding_label`. Provenance (DOI, source line, URL) is complete except the two homology-modeled BoNT/A2 sets and the Jin rows. `transcription_confidence` is uniformly `high` — treat as a soft flag, possibly a default fill. `[verify]` spot-check 1–2 known hotspots against the source papers.

Clean MD-correlation core. Dropping homology models (`HM_`) and EC50/excluded rows leaves **163 rows / 5 antigens on real PDBs** (VEGF, lysozyme, HPr, BoNT/A1, MT-SP1).

4 First system: hen egg-white lysozyme (HEL)

HEL is the cleanest entry point: small (129 residues), rigid (apo ≈ bound), very stable, textbook AMBER behavior, and — crucially — it carries **three** independent antibody alanine scans in the benchmark, on a consistent 1–129 numbering:

Antibody	PDB	Ag chain	# Ala	$\Delta\Delta G$ range (kcal/mol)
HyHEL-63	1DQJ	C	11	0.60–6.10
D1.3	1VFB	C	12	0.20–2.90
HyHEL-10	3HFM	Y	4	1.03–6.83

Why this is better than one antibody. HyHEL-10 and HyHEL-63 share an epitope and *agree* on hotspots (K96: 6.83 vs 6.10; Y20: 4.87 vs 3.30; K97: 6.18 vs 3.50; R21: 1.03 vs 1.30) — direct evidence for the antigen-intrinsic hotspot idea (Design note 1). D1.3 binds a *non-overlapping* epitope (residues 18–24 and 116–129; zero residue overlap with the HyHELs), giving a second epitope plus built-in negative controls. Because we extract features from the apo antigen, **one apo-HEL trajectory** serves all three antibodies and their mean.

Starting structure. RCSB **1AKI** (1.5 Å X-ray, chain A, apo). Verified: the 1AKI sequence matches the benchmark WT identity at all 23 labeled positions, so labels map 1:1 onto 1–129 numbering (K96/K97/Y20 hotspots confirmed). Apo crystal chosen over antigen-stripped-from-complex to match the prediction scenario; HEL's rigidity makes the choice low-risk.

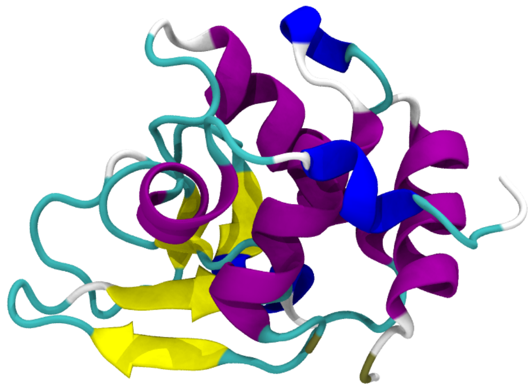


Figure 1: Apo HEL starting structure (RCSB **1AKI**, 1.5 Å X-ray, chain A); cartoon coloured by secondary structure (helices purple, sheets yellow). This single apo trajectory serves all three antibody scans.

5 Simulated antigens

One section per antigen we simulate: what it is, the structural handle we use, and its alanine-scan labels. Each structure is coloured per residue by $\Delta\Delta G_{\text{bind}}$ on a diverging scale with **white pinned at** 0: red = positive (hotspot / likely epitope), blue = negative, grey = residue not scanned. One panel per antibody; a single global colour scale ($\pm\max|\Delta\Delta G_{\text{bind}}|$ over the whole benchmark, white at 0) is shared by every figure, so colours are directly comparable across antigens. Vertical colourbar at right. $\Delta\Delta G_{\text{bind}}$ is the change in binding free energy, K_d/K_i-derived; positive weakens binding.

5.1 Lysozyme (hen egg-white, HEL)

Hen egg-white lysozyme is a 129-residue glycoside hydrolase and the canonical model antigen of structural immunology: multiple high-resolution antibody complexes and alanine scans exist (HyHEL-10, HyHEL-63, D1.3). It is compact and conformationally rigid, cross-linked by four disulfides. We simulate the apo enzyme from `1AKI`. The labels here are the HyHEL-63 scan (`1DQJ`, Table 1) [7]; the v2 benchmark adds D1.3 (`1VFB`) [8] and HyHEL-10 (`3HFM`) [9] scans for the same antigen. These $\Delta\Delta G_{\text{bind}}$ values are drawn from the AB-Bind database [4].

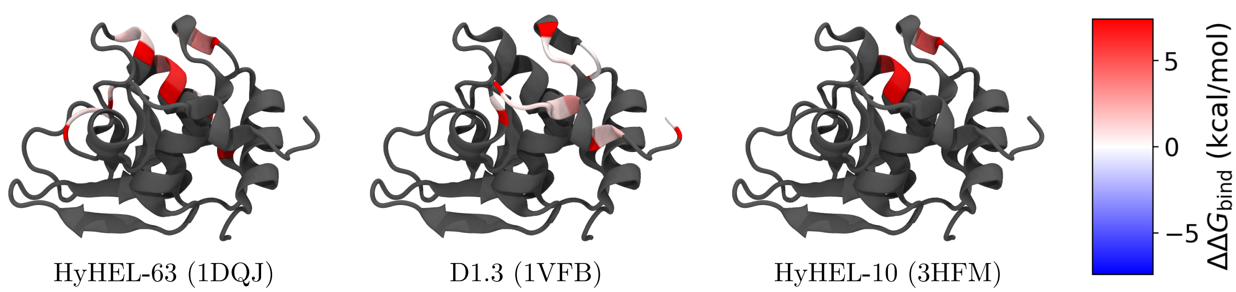


Figure 2: HEL coloured per residue by alanine-scan $\Delta\Delta G_{\text{bind}}$ for its three antibodies (shared scale, white = 0, red = hotspot, blue < 0; grey = not scanned). Distinct but overlapping epitope cores across HyHEL-63, D1.3 and HyHEL-10.

Table 1: **Lysozyme** alanine-scan $\Delta\Delta G_{\mathrm{bind}}$ (kcal/mol; SPR, K_d-derived via $\Delta\Delta G_{\mathrm{bind}} = RT\ln(K^{\mathrm{mut}}/K^{\mathrm{wt}})$, $T = 310\,\mathrm{K}$; start structure: apo 1AKI). Positive $\Delta\Delta G_{\mathrm{bind}}$ weakens binding.

Residue	HyHEL-63 (1DQJ)	IgG1-kappa D1.3 Fv (1VFB)	HyHEL-10 Fv (3HFM)
D18	–	+0.30	–
N19	–	+0.30	–
Y20	+3.30	–	+4.87
R21	+1.30	–	+1.03
Y23	–	+0.40	–
S24	–	+0.80	–
W62	+0.80	–	–
W63	+1.30	–	–
L75	+1.50	–	–
T89	+0.80	–	–
N93	+0.60	–	–
K96	+6.10	–	+6.83
K97	+3.50	–	+6.18
S100	+0.80	–	–
D101	+1.30	–	–
K116	–	+0.70	–
T118	–	+0.80	–
D119	–	+1.00	–
V120	–	+0.90	–
Q121	–	+2.90	–
I124	–	+1.20	–
R125	–	+1.80	–
L129	–	+0.20	–

5.2 HPr

The histidine-containing phosphocarrier protein (HPr) is a small (~9 kDa, 85-residue) cytoplasmic protein of the bacterial phosphoenolpyruvate:sugar phosphotransferase system, shuttling a phosphoryl group between enzyme I and the sugar-specific EIIA components. The *E. coli* protein is a well-folded open-faced β-sandwich with *no cysteines*, making it an exceptionally clean dynamics test case. The Jel42 epitope clusters in the C-terminal half (Table 2) [10]; $\Delta\Delta G_{\mathrm{bind}}$ values via AB-Bind [4]. We simulate the apo protein from `2JEL` (chain P).

Jel42 (2JEL)

Figure 3: HPr coloured per residue by Jel42 alanine-scan $\Delta\Delta G_{\mathrm{bind}}$ (white = 0, red = hotspot; grey = not scanned). A single dominant hotspot (E70).

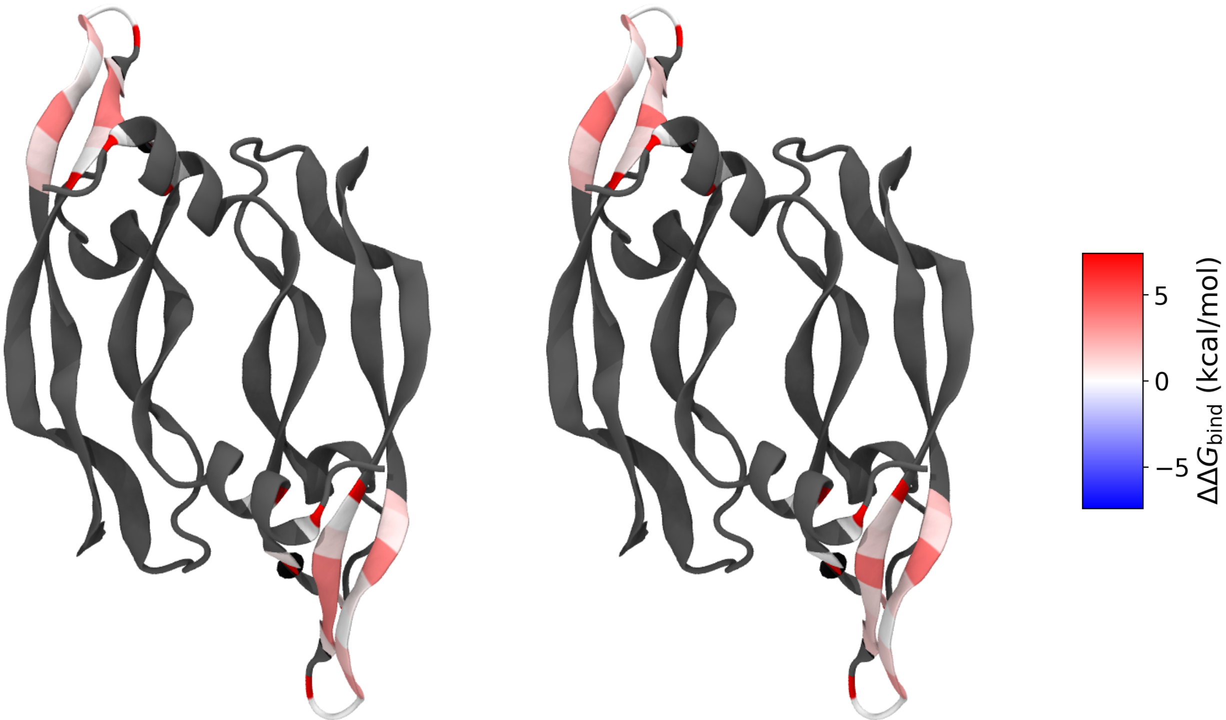
Table 2: **HPr** alanine-scan $\Delta\Delta G_{\mathrm{bind}}$ (kcal/mol; competition, K_d-derived via $\Delta\Delta G_{\mathrm{bind}} = RT\ln(K^{\mathrm{mut}}/K^{\mathrm{wt}})$, $T = 310\,\mathrm{K}$; start structure: strip 2JEL chain P). Positive $\Delta\Delta G_{\mathrm{bind}}$ weakens binding.

Residue	Jel42 (2JEL)
T62	+0.00
E68	+0.41
E70	+2.75
H76	-0.41
E83	+0.00
E85	+0.00

5.3 VEGF-A

Vascular endothelial growth factor A is a secreted pro-angiogenic cytokine and a central drug target (bevacizumab, ranibizumab). Its receptor-binding domain is an antiparallel, disulfide-linked homodimer with a cystine-knot growth-factor fold; the two protomers are covalently joined, so the biological unit must be simulated as the dimer. The benchmark pairs two antibodies against the same receptor-binding face: fab-12 (`1BJ1`, the bevacizumab precursor) [11] and the affinity-matured

Y0317 (`1CZ8`, ranibizumab) [12] — a useful same-antigen/different-affinity comparison (Table 3); $\Delta\Delta G_{\mathrm{bind}}$ values via AB-Bind [4]. We simulate the apo dimer from `1BJ1` (chains V, W).



fab-12 (1BJ1) Y0317 (1CZ8)

Figure 4: VEGF-A homodimer coloured per residue by $\Delta\Delta G_{\mathrm{bind}}$ for each antibody (shared scale, white = 0; grey = not scanned). Both protomers carry the same labelled residues; red marks the hotspots.

5.4 MT-SP1 (matriptase)

MT-SP1 / matriptase (ST14) is a type II transmembrane serine protease of the S1 family; its extracellular catalytic domain is over-expressed in epithelial cancers and is a validated oncology target. The benchmark uses two inhibitory Fabs, E2 (`3BN9`) [13] and S4 (`3NPS`) [14], scanning the protease domain in chymotrypsin numbering (Table 4; $\Delta\Delta G_{\mathrm{bind}}$ via AB-Bind [4]). Simulation is staged but **pending**: the crystal structures lack two internal residues (149, 218) that must be repaired before MD (§9).

`figures/mtsp1_structure.png`
(not yet regenerated; see figures/FIGURES.md)

Figure 5: MT-SP1 protease domain (3NPS chain A). *Placeholder.*

Table 3: **VEGF** alanine-scan $\Delta\Delta G_{\text{bind}}$ (kcal/mol; SPR, K_d-derived via $\Delta\Delta G_{\text{bind}} = RT\ln(K^{\text{mut}}/K^{\text{wt}})$, $T = 310\,\text{K}$; start structure: strip 1BJ1 chains V,W (dimer)). Positive $\Delta\Delta G_{\text{bind}}$ weakens binding.

Residue	fab-12 (1BJ1)	Y0317 (1CZ8)
F17	+0.00	+0.00
Y21	+0.00	+0.00
Y45	+0.82	+1.94
K48	+0.41	+0.00
Q79	+0.00	+0.65
I80	+0.82	+0.95
M81	+3.69	+4.10
R82	+3.69	+0.82
I83	+3.69	+1.30
K84	+0.65	+1.36
H86	+0.00	+0.00
Q87	+0.00	+0.00
G88	+2.76	+2.67
Q89	+1.75	+1.06
H90	+0.00	+0.00
I91	+0.41	+1.06
G92	+3.69	+4.10
E93	+0.82	+1.15
M94	+1.42	+1.91

5.5 Bont/A1 (H_C receptor-binding domain)

Botulinum neurotoxin serotype A1 is among the most potent toxins known: a ~150 kDa di-chain protein comprising a zinc-metalloprotease light chain and a heavy chain. Its C-terminal heavy-chain domain (H_C, here residues 872–1295) is the receptor-binding domain — engaging neuronal gangliosides and the SV2 protein — and the principal protective antigen targeted by neutralizing antibodies. We simulate the isolated H_C domain (`2NYY` chain A); the catalytic light-chain zinc lies outside this fragment. Two neutralizing mAbs, CR1 (`2NYY`) and AR2 (`2NZ9`), provide the labels (Table 5) [15]; $\Delta\Delta G_{\text{bind}}$ via AB-Bind [4].

Table 4: **MT-SP1** alanine-scan $\Delta\Delta G_{\mathrm{bind}}$ (kcal/mol; enzymatic inhibition, K_i-derived via $\Delta\Delta G_{\mathrm{bind}} = RT\ln(K^{\mathrm{mut}}/K^{\mathrm{wt}})$, $T = 310\,\mathrm{K}$; start structure: strip 3NPS chain A). Positive $\Delta\Delta G_{\mathrm{bind}}$ weakens binding.

Residue	E2 (3BN9)	S4 (3NPS)
Q23	–	+0.03
I26	–	+0.65
Q38	-0.42	–
I41	+0.00	–
I45	–	-0.34
D46	–	+0.34
D47	–	+1.08
R48	–	-1.07
F50	–	+0.22
R51	–	+0.14
Y52	–	+0.46
I60	+0.84	–
R82	–	-0.15
R87	-0.16	–
F89	–	+1.61
N90	–	+0.26
D91	–	+1.52
F92	–	+0.47
T93	–	+0.73
F94	+0.64	–
N95	+0.77	–
D96	+6.70	–
F97	+6.70	–
T98	+1.13	–
H138	–	+1.89
Q140	–	+0.30
Y141	–	+1.79
H143	+0.09	–
T144	–	+0.18
Q145	+0.13	–
Y146	+1.09	–
L147	–	+0.30
T150	+0.29	–
L153	+0.34	–
E163	–	+0.62
Q168	–	-0.06
E169	+0.37	+0.75
Q174	-0.03	–
Q175	+2.51	–
D214	–	+1.48
D217	+0.57	–
Q218	–	-0.04
R219	–	-0.08
K221	–	-0.10
R222	-0.09	–
K224	+0.79	–

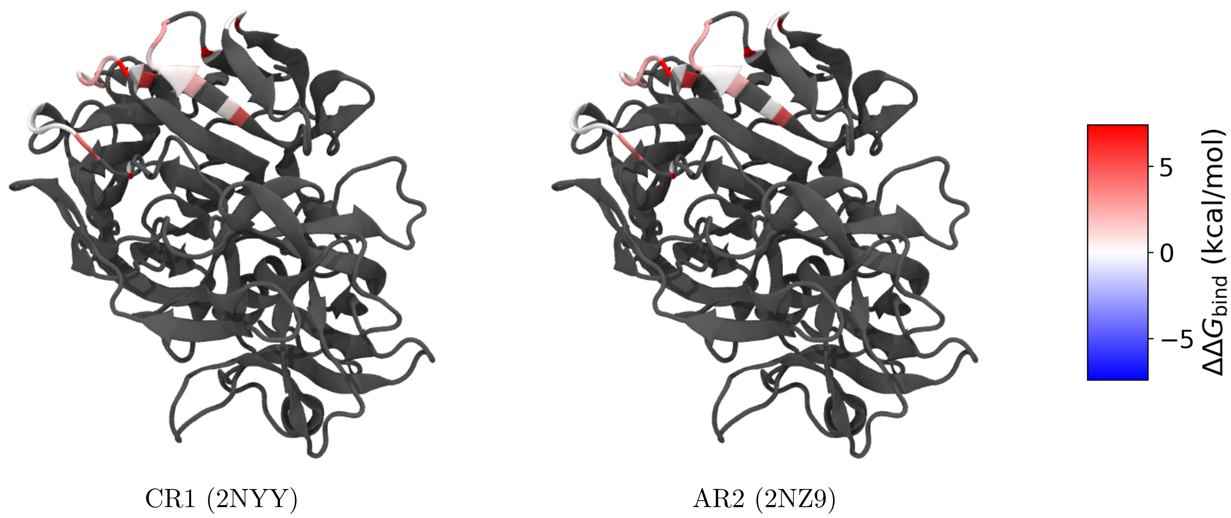


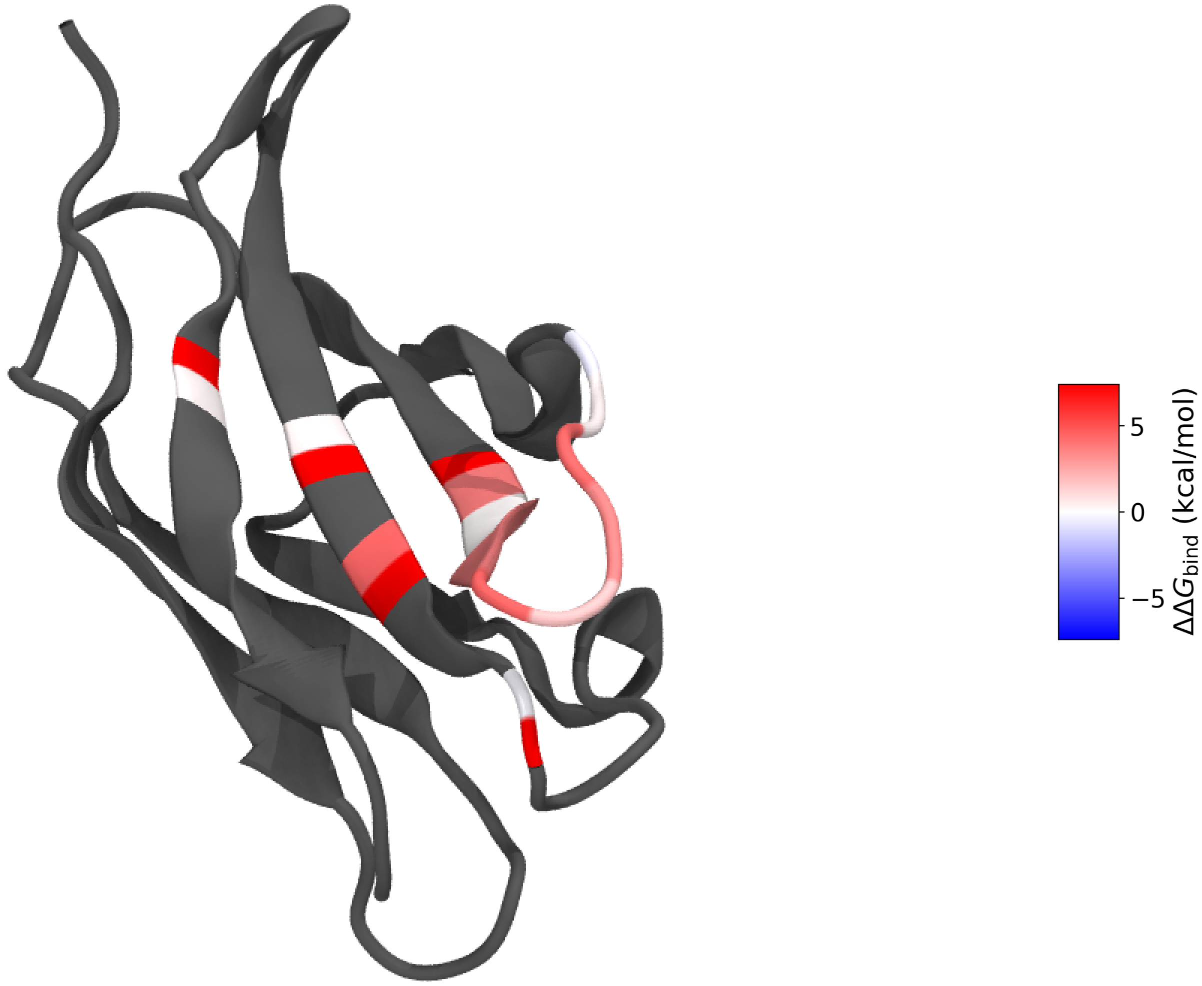
Figure 6: Bont/A1 H_C coloured per residue by $\Delta\Delta G_{\text{bind}}$ for two neutralizing mAbs (shared scale, white = 0; grey = not scanned). Red marks the hotspots — the strongest reach $\Delta\Delta G_{\text{bind}} > 7$ (binding essentially abolished).

Table 5: **Bont/A1** alanine-scan $\Delta\Delta G_{\text{bind}}$ (kcal/mol; KinExA, K_d-derived via $\Delta\Delta G_{\text{bind}} = RT\ln(K^{\text{mut}}/K^{\text{wt}})$, $T = 310\,\text{K}$; start structure: extract 2NYY Hc (res 872-1295)). Positive $\Delta\Delta G_{\text{bind}}$ weakens binding.

Residue	CR1 (2NYY)	AR2 (2NZ9)
S902	-0.23	-0.12
K903	+0.29	+0.54
Q915	+0.52	+0.10
F917	+0.42	-0.05
N918	+0.89	+2.16
L919	+2.59	+2.28
E920	+2.84	+2.77
K923	-0.14	-0.30
F953	+4.06	+3.34
N954	+0.10	-0.15
S955	+0.00	-0.08
I956	-0.01	+0.07
K1056	-0.03	-0.01
D1058	+0.01	-0.03
R1061	+0.82	+0.29
D1062	+2.53	+2.34
T1063	+1.62	+2.37
H1064	+7.32	+7.42
R1294	+0.30	+0.39

5.6 IFN-γ receptor

The interferon-γ receptor α-chain (IFN-γR1) ectodomain is the high-affinity ligand-binding subunit of the type II interferon receptor: two fibronectin type III domains forming an L-shaped β-sandwich that engages the IFN-γ homodimer. The neutralizing antibody A6 binds this ectodomain and the published alanine scan localises a compact energetic hotspot on one face (Table 6) [16]; $\Delta\Delta G_{\mathrm{bind}}$ values curated via SKEMPI 2.0 [6]. It is a clean, rigid, single-chain ~95-residue Ig-like domain — an attractive small test antigen. We simulate the apo ectodomain stripped from `1JRH` (chain I).



A6 (1JRH)

Figure 7: IFN-γ receptor coloured per residue by A6 alanine-scan $\Delta\Delta G_{\mathrm{bind}}$ (shared scale, white = 0, red = hotspot; grey = not scanned). The hotspots cluster on a single β-sandwich face.

5.7 Tissue factor

Tissue factor (TF, coagulation factor III) is the cell-surface initiator of the extrinsic blood-coagulation cascade: its extracellular region is a two-domain fibronectin type III module that binds and allosterically activates factor VIIa. The inhibitory Fab 5G9 blocks the macromolecular substrate site, and its antigen-side alanine scan marks the functional epitope (Table 7) [17]; $\Delta\Delta G_{\mathrm{bind}}$ values curated via SKEMPI 2.0 [6]. The ectodomain is a well-folded, single-chain ~200-residue β-rich protein. We simulate the apo ectodomain stripped from `1AHW` (chain C).

Table 6: **IFN-gamma receptor** alanine-scan $\Delta\Delta G_{\text{bind}}$ (kcal/mol; SPR, K_d-derived via $\Delta\Delta G_{\text{bind}} = RT\ln(K^{\text{mut}}/K^{\text{wt}})$, $T = 310\,\text{K}$; start structure: strip 1JRH chain I). Positive $\Delta\Delta G_{\text{bind}}$ weakens binding.

Residue	A6 (1JRH)
K47	+3.71
N48	+0.17
Y49	+3.53
G50	+4.45
V51	+1.68
K52	+3.39
N53	+4.30
S54	+0.38
E55	-0.57
N79	-0.20
W82	+4.43
R84	+0.39
K98	+0.31

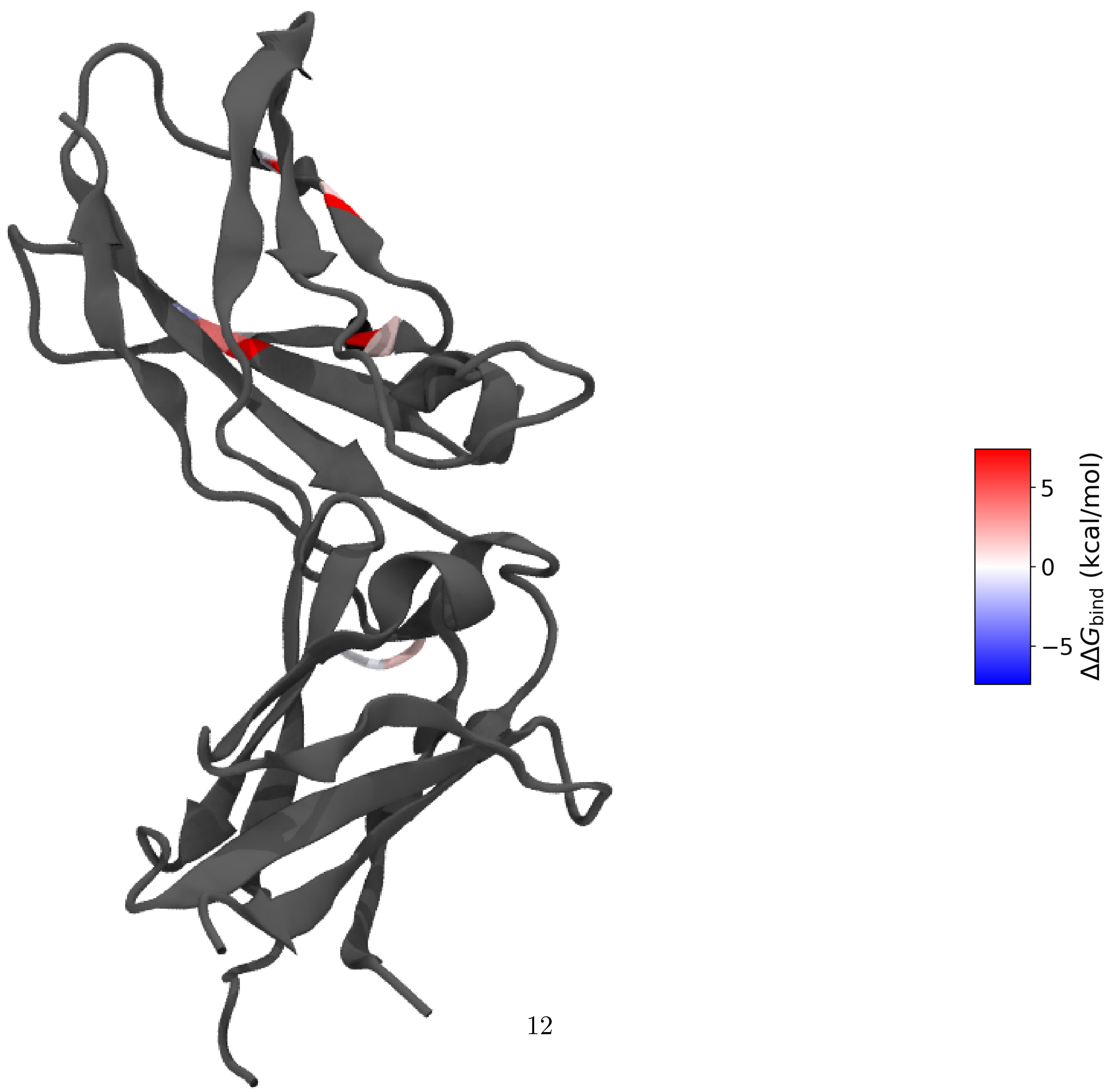
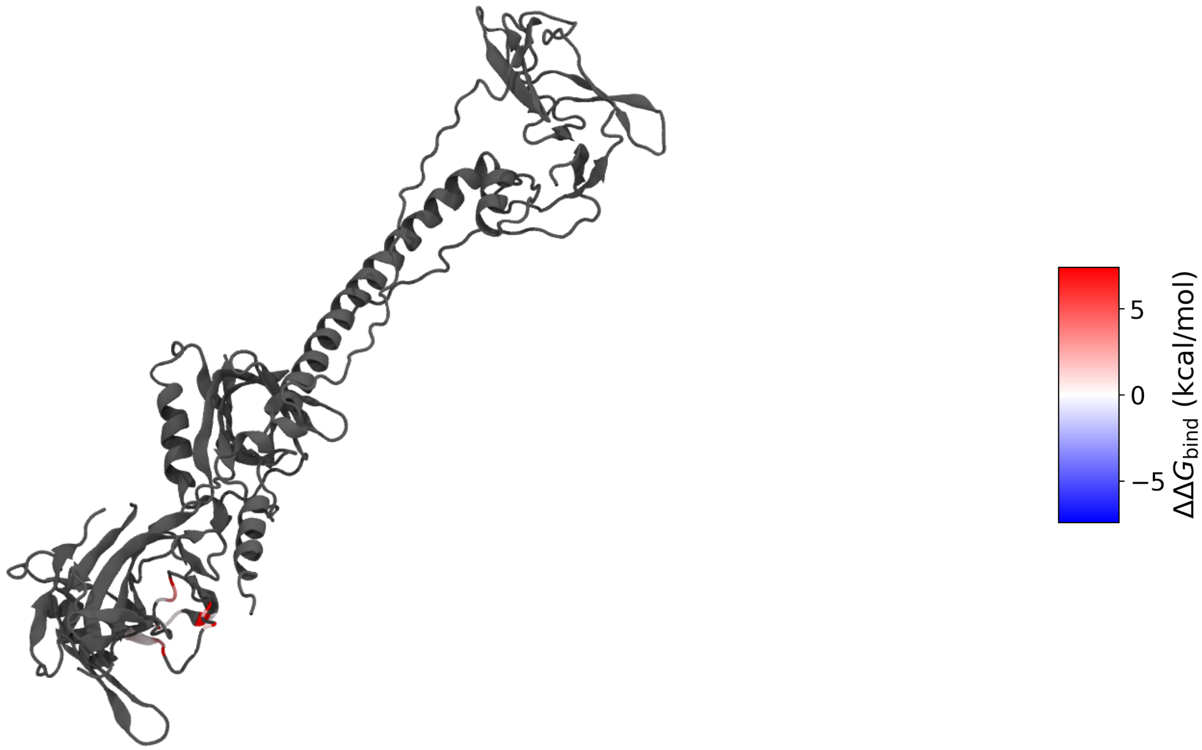


Table 7: **Tissue factor** alanine-scan $\Delta\Delta G_{\text{bind}}$ (kcal/mol; SPR, K_d-derived via $\Delta\Delta G_{\text{bind}} = RT\ln(K^{\text{mut}}/K^{\text{wt}})$, $T = 310\,\text{K}$; start structure: strip 1AHW chain C). Positive $\Delta\Delta G_{\text{bind}}$ weakens binding.

Residue	5G9 (1AHW)
Y156	+3.81
Y157	-1.89
T167	-0.07
T170	+1.11
L176	+0.99
D178	-0.48
T197	+1.35
V198	-0.31
N199	+1.08

5.8 HCMV glycoprotein B

Glycoprotein B (gB) is the conserved class III fusion machine of human cytomegalovirus and a principal target of neutralizing antibodies. The ectodomain is a large, elongated trimer; the antibody 1G2 recognises antigenic domain 2 and its alanine scan identifies the contact hotspots (Table 8) [18]; $\Delta\Delta G_{\text{bind}}$ values curated via SKEMPI 2.0 [6]. This is our largest antigen (∼566 modelled residues) and is biologically trimeric — we simulate the apo protomer as deposited in `5C6T` (chain A), noting that the native oligomeric state may modulate the dynamics of the buried interfaces.



1G2 (5C6T)

Figure 9: HCMV glycoprotein B protomer coloured per residue by 1G2 alanine-scan $\Delta\Delta G_{\text{bind}}$ (shared scale, white = 0, red = hotspot; grey = not scanned).

Table 8: **HCMV glycoprotein B** alanine-scan $\Delta\Delta G_{\text{bind}}$ (kcal/mol; SPR, K_d-derived via $\Delta\Delta G_{\text{bind}} = RT\ln(K^{\text{mut}}/K^{\text{wt}})$, $T = 310\,\text{K}$; start structure: strip 5C6T chain A). Positive $\Delta\Delta G_{\text{bind}}$ weakens binding.

Residue	1G2 (5C6T)
Y280	+3.21
N281	+0.28
T283	+0.34
N284	+0.36
N286	-0.37
F290	+2.00
E292	+1.09
F297	+2.76
F298	+0.41
I299	+0.89

5.9 Human growth hormone

Human growth hormone (hGH, somatotropin) is a 191-residue four-helix-bundle cytokine and the historical birthplace of alanine-scanning mutagenesis. Here the antibody-specific labels are the

MAb3 scan (Table 9) [5] **[verify]** (benchmark cites L. Jin's 1994 thesis, Table 3.2; the published values are in [5]); the benchmark additionally carries a secondary, provisional profile of the total $\Delta\Delta G_{\text{bind}}$ aggregated across 21 anti-hGH MAbs (not antibody-specific, held out of the primary labels). We simulate the apo hormone from `1HGU` (chain A).

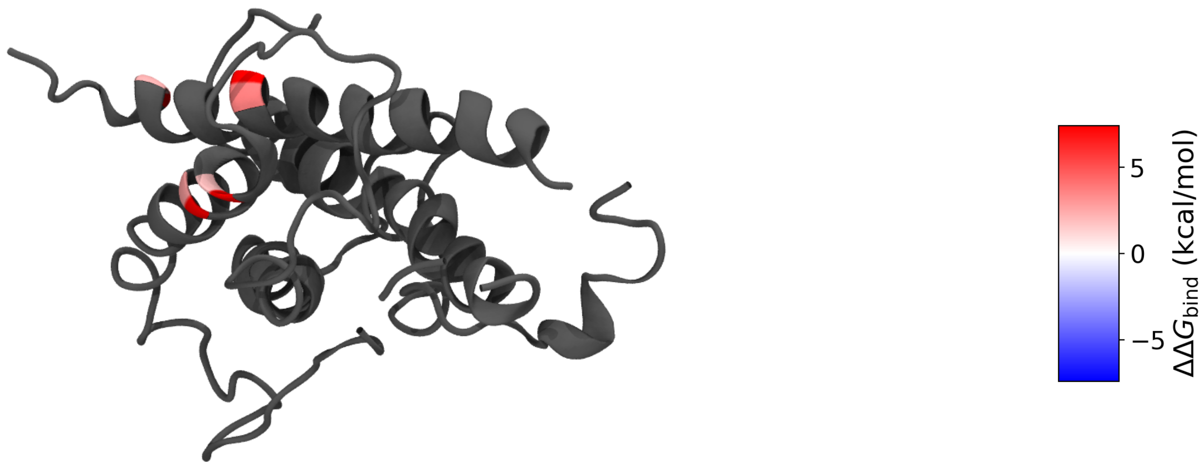


Figure 10: Human growth hormone coloured per residue by MAb3 alanine-scan $\Delta\Delta G_{\text{bind}}$ (shared scale, white = 0, red = hotspot; grey = not scanned). Only the four MAb3-scanned positions are coloured.

Table 9: **human growth hormone** alanine-scan $\Delta\Delta G_{\text{bind}}$ (kcal/mol; ELISA, K_d-derived via $\Delta\Delta G_{\text{bind}} = RT\ln(K^{\text{mut}}/K^{\text{wt}})$, $T = 310\,\text{K}$; start structure: apo 1HGU). Positive $\Delta\Delta G_{\text{bind}}$ weakens binding.

Residue	MAb3 (1HGU)
R8	+2.16
R16	+3.68
D112	+1.84
D116	+2.73

6 A second label tier: binary critical-residue maps

The $\Delta\Delta G_{\text{bind}}$ antigens above are the gold standard — a graded, energetic importance per residue — but experimental alanine scans with full $\Delta\Delta G_{\text{bind}}$ tables are scarce. A much larger body of data exists at a coarser but still *energetic* resolution: **saturating shotgun alanine scans** read out by flow cytometry (Integral Molecular's platform; Davidson & Doranz [19]), which classify each scanned residue as *critical* or *non-critical* for a given antibody. A residue is critical when its mutation measurably abolishes binding, so even this binary call is grounded in binding energy, not mere proximity — it is the same quantity as $\Delta\Delta G_{\text{bind}}$, thresholded. We therefore treat these as a second label tier: **Tier 1**, graded $\Delta\Delta G_{\text{bind}}$ (§5); **Tier 2**, binary critical/non-critical.

Source and the proximity filter. We mine these from IEDB (`query-api.iedb.org`) [1], keeping only conformational (discontinuous) epitopes defined by *binding* assays and explicitly discarding every structure/contact-derived record — the proximity labels this project argues against (recall that $> 99\%$ of structure-linked IEDB epitopes are crystal/cryo-EM contact, §1). Within the binding-derived set we take the flow-cytometry (shotgun-mutagenesis) subset, whose curated residue list *is* the set of critical residues. This yields 3,213 antibody–antigen epitope maps over 186 antigens. Table 10 lists the simulatable, not-yet-included candidates; note the depth of antibody sampling per antigen (e.g. dengue envelope: 781 mapped antibodies across 30 studies), which lets a Tier-2 antigen carry a *consensus* epitope (residues critical across many antibodies) as well as antibody-specific maps.

Table 10: **Tier-2 (binary) label candidates** from IEDB shotgun-mutagenesis alanine scans (flow-cytometry critical-residue mapping; Integral Molecular / Doranz). Labels are per-residue *critical vs non-critical*, energetically grounded (a critical residue is one whose mutation measurably weakens antibody binding) but coarser than $\Delta\Delta G_{\text{bind}}$. "Antibodies" = mapped antibody footprints (one antibody → one conformational epitope); "Studies" = distinct source papers; "Crit. res." = union of critical residues across all antibodies. Structures provisional ([v] = numbering to be verified before use).

Antigen	Organism	Antibodies	Studies	Crit. res.	Structural handle
Dengue E (envelope)	Dengue virus	781	30	632	sE ectodomain, e.g. 1OKE [v]
Chikungunya E1/E2	Alphavirus chikungunya	128	6	98	ectodomain, e.g. 3N42 [v]
RVFV Gn	Rift Valley fever virus	63	8	210	Gn ectodomain, e.g. 6F9F [v]
Influenza HA	Influenza A virus	81	22	296	HA ectodomain (trimeric) [v]
GRO-α / CXCL1	Homo sapiens (human)	30	1	23	chemokine, e.g. 1MGS [v]
CD4	Homo sapiens (human)	1	1	18	D1–D2, e.g. 1CDH [v]
CD40L	Homo sapiens (human)	1	1	13	TNF domain, e.g. 1ALY [v]
IL-2Rα	Homo sapiens (human)	2	1	13	ectodomain, e.g. 1Z92 [v]
IL-7Rα	Homo sapiens (human)	2	1	32	ectodomain, e.g. 3DI2 [v]
HLA class I	Homo sapiens (human)	15	5	38	peptide-MHC, many [v]

Numbering verification. The shotgun residues are reported in UniProt numbering and carry their wild-type identity, but aggregating by UniProt accession mixes proteins, numbering conventions, and (for viruses) serotypes/subtypes. We therefore map every residue onto a single reference structure and *discard any residue whose wild-type identity does not match* — exactly the wild-type check applied to the $\Delta\Delta G_{\text{bind}}$ antigens. This is strict: of dengue's raw critical positions only those matching DENV2 envelope survive, and likewise only H3 residues survive for influenza. What remains is clean and biologically coherent — the top dengue residue (fusion loop, E101) is called critical by 102 of 632 antibodies.

Representation. We render each staged antigen two ways (Figs. 11, 12): **binary** (red = critical for at least one antibody, grey = not) and **consensus frequency** (white→red by the number of antibodies for which the residue is critical). The frequency view is itself an energetic-importance signal — it ranks residues by immunodominance across hundreds of antibodies — and reduces to the binary view under any threshold, so we keep both and decide later. Tier-2 antigens enter the benchmark as a separate (binary/graded) label stream, never merged into the $\Delta\Delta G_{\text{bind}}$ tables; in the leakage-controlled evaluation they are held out by antigen.

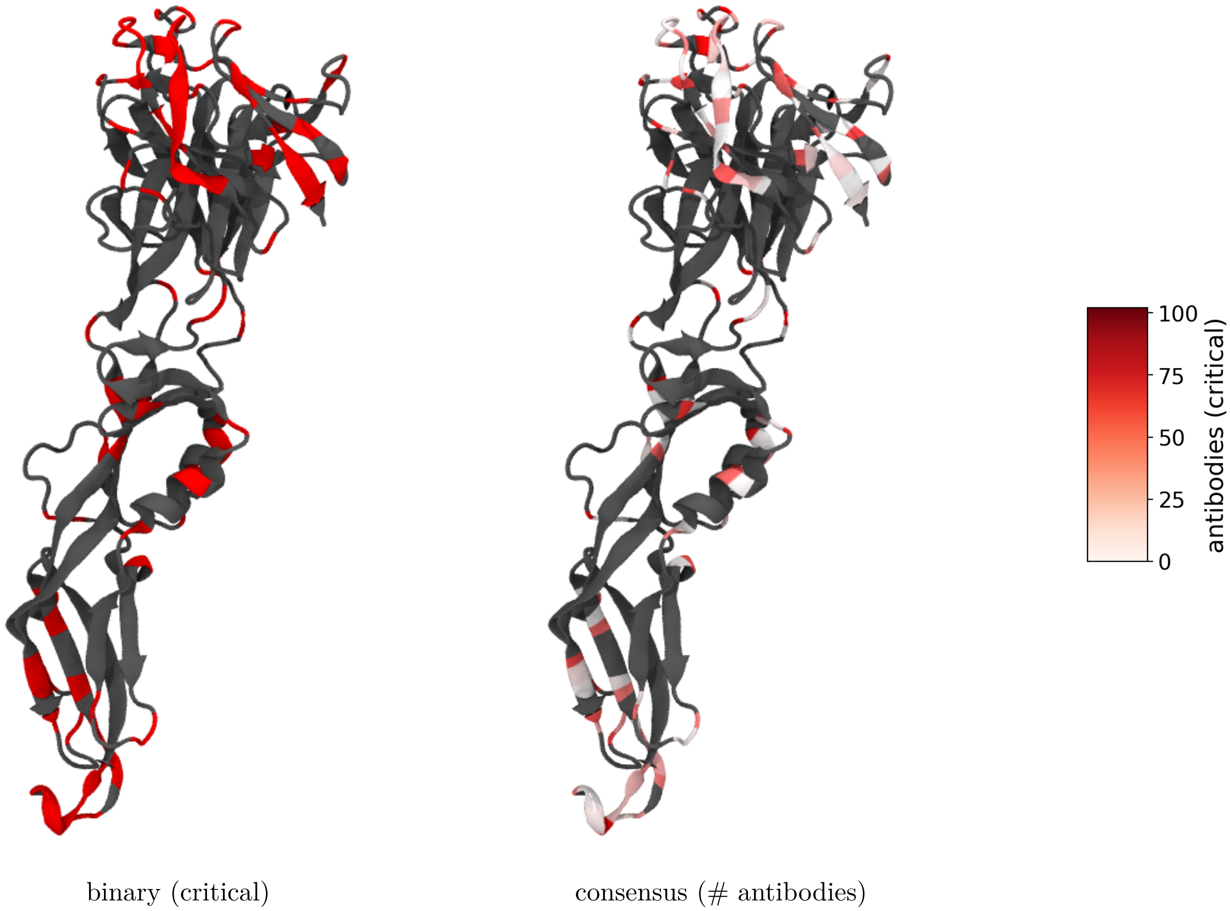


Figure 11: **Dengue envelope (DENV2, `1OKE` sE):** 96 wild-type-verified critical residues from 632 antibodies. Binary (left) and antibody-frequency consensus (right). The deepest red is the conserved fusion loop (E101, critical for 102 antibodies); a second focus is the EDIII lateral ridge — both textbook dengue neutralization sites.

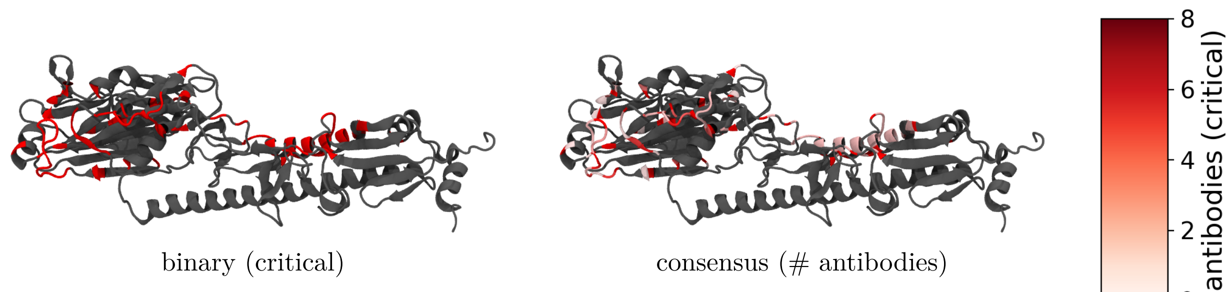


Figure 12: **Influenza hemagglutinin (H3 X-31, `1HGG` protomer):** 84 wild-type-verified critical residues (HA1 head + HA2 stem) from 79 antibodies, binary and antibody-frequency consensus. Critical residues concentrate on the HA1 globular head (the classic antigenic sites) with a sparser stem contribution. Native HA is an obligate trimer; one protomer is shown.

7 The static-exposure baseline

Before adding any dynamics, we ask how far a purely *static* surface descriptor already gets — both to set the bar the dynamics features must clear and to make the proximity/exposure critique of §1 quantitative on our own labels. The descriptor is per-residue **relative solvent-accessible surface area** (rel. SASA): Shrake–Rupley SASA on the apo structure, normalised by the residue-type maximum [20]. This is a single-frame calculation on the apo input structures — the pre-trajectory baseline; once the production runs land it is replaced by trajectory-averaged SASA at the same residue keys (and joined by RMSF, non-bonded energetics, and hydration).

Setup. We pool every residue of the eight $\Delta\Delta G_{\text{bind}}$ antigens (1,790 residues). A residue is *important* if its alanine-scan $\Delta\Delta G_{\text{bind}} \geq 1\,\text{kcal/mol}$ for any antibody (aggregated as the max over antibodies). An exposure-only predictor labels a residue an epitope if rel. SASA ≥ 0.25 (the standard “exposed” cutoff) — the static analogue of a proximity label. We then read off precision and recall against the experimental importance labels.

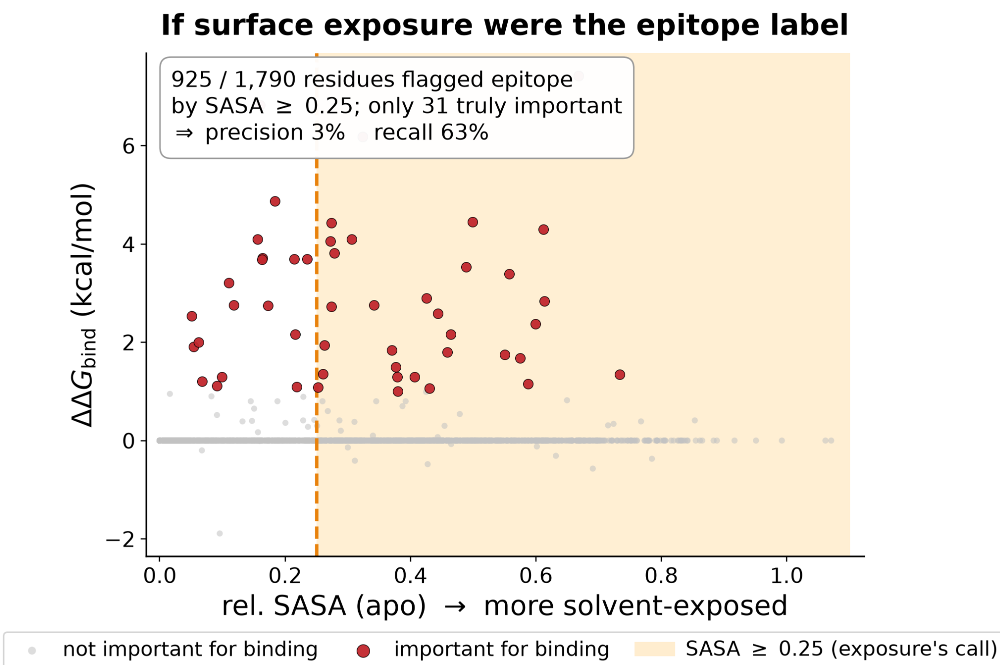


Figure 13: **Surface exposure as an epitope label.** Every residue of the eight $\Delta\Delta G_{\text{bind}}$ antigens, apo rel. SASA (x) vs. alanine-scan $\Delta\Delta G_{\text{bind}}$ (y); red = experimentally important for binding, grey = not; the shaded band is what an exposure cutoff (rel. SASA ≥ 0.25) would call epitope. Exposure flags 925 of 1,790 residues as epitope but only 31 are truly important (precision $\approx 3\%$), while still missing the buried-at-interface hotspots to the left of the line (recall $\approx 63\%$). Single-frame apo SASA; `analysis/compute_sasa.py` + `plot_feature_vs_label.py`.

Result. Exposure behaves exactly as the thesis predicts. It has *moderate recall* (it knows roughly where the surface is: 63% of importance hotspots are exposed) but *near-zero precision* ($\approx 3\%$): rel. SASA ≥ 0.25 sweeps in almost half the protein, overwhelmingly residues that contribute no binding energy. Per antigen the precision is 0–14% (highest for the clean single-domain antigens `1AKI`, `1BJ1`, `1JRH`). And *among the scanned interface residues*, exposure does not even rank graded importance: the pooled Spearman correlation between rel. SASA and $\Delta\Delta G_{\text{bind}}$ is -0.16 — once a residue is on the surface, how exposed it is says nothing about how much binding energy it carries. That gap — high recall, no precision, no within-interface ranking — is precisely what the dynamic and energetic features are meant to fill, and every feature we add is scored against this baseline.

Caveats. (i) Single-frame apo SASA; trajectory-averaging will move values and is the real control. (ii) `5C6T` (gB) is anomalous here — it is scored on the unrepaired structure (§9) *and* as an isolated protomer of a native trimer, so residues buried in the trimer interface read as exposed; `2JEL` (HPr) carries only one hotspot. (iii) Residues never scanned are treated as non-epitope, which alanine scans largely justify (they target the known epitope) but which inflates the negative set.

8 Methods — apo-HEL run #1

Frozen, signed-off stack in `../docs/METHODS.md` (2026-06-18). Purpose of run #1: *validate the prep→equil→production pipeline on Gemini and extract a first RMSF / flexibility feature* (the Westhof re-test). House default stack signed off as the starting point.

Stack. AMBER99SB-ILDN / TIP3P; cubic box, 1.2 nm clearance; neutralize + 0.15 M NaCl; leapfrog, 2 fs, LINCS on h-bonds; PME ($r_{\text{coul}} = r_{\text{vdw}} = 1.0$ nm, potential-shift, DispCorr); V-rescale at 300 K (Protein / Non-Protein); EM → 100 ps NVT → 100 ps NPT (both position-restrained) → **100 ns** production (single trajectory, for RMSF statistics). Random seeds (`gen_seed` = `ld_seed` = -1): bitwise reproducibility is not the target — the workflow reproduces from git-tracked inputs and saved derived outputs, and random seeds give genuinely independent replicas. Frozen per-run `.mdp` files: `md/1AKI/apo/configs/` (run dirs are organized `md/<PDB>/<state>/`).

Deliberate deviations from the house scripts. (i) **C-rescale** barostat for NPT equilibration (Parrinello-Rahman only in production), since PR can ring when started far from equilibrium; (ii) **cubic** box (the old house scripts disagreed cubic/dodecahedron) — chosen for interpretability and uniform across all systems, at a modest extra-solvent cost vs. dodecahedron.

HEL-specific checkpoints (not glossed over). pdb2gmx must form HEL's **4 disulfides** (6–127, 30–115, 64–80, 76–94) — verify the log reports 4 SS bonds. His15 protonation: record the assigned state. Crystal waters (78 HOH) removed; fresh TIP3P added. Read every grompp warning rather than letting `-maxwarn` hide it.

Caveats / flags. TIP3P over-mobilizes water and 2 fs+LINCS suppress H dynamics — fine for relative RMSF and pipeline validation, *not* for the dynamics-grade features. 100 ns is a single trajectory for statistics; replicas are deferred (random seeds make adding independent ones trivial).

Deferred: the distinctive run. FRESEAN mode participation and 3D-2PT hydration thermodynamics require a *separate* protocol — high-frequency velocity output, constraint removal / sub-fs dt, TIP4P/2005 — with settings pulled from the FRESEAN and energy-transfer manuscripts. This is the run that powers the mode-based and hydration feature classes.

9 Structure completeness and the repair strategy

Several crystal structures are incomplete — disordered termini, internal loops, or whole segments that the experiment never resolved (PDB `REMARK 465`). Apo-MD demands a chemically complete chain, so each gap forces a choice. We make that choice explicitly per structure rather than feeding everything through one tool, because **the right repair scales with gap length**, and the wrong tool on a long gap produces a confident-looking but meaningless model.

What a "fixer" can and cannot supply. The instinct to "just grab the FASTA for the missing residues" conflates two different problems. The deposited sequence (`SEQRES`, the same residues a fixer recovers automatically by diffing `SEQRES` against the resolved `ATOM` records) supplies the *identity* of the missing residues — but identity is the easy half. The hard half is *geometry*: where the missing backbone atoms sit in three dimensions, about which the sequence says nothing. `pdbfixer` (OpenMM) builds missing heavy atoms onto resolved residues from templates — valid and routine — and threads missing residues in with default internal coordinates followed by a short minimization of only the new atoms. That minimization removes clashes; it performs *no* conformational sampling. The result is therefore reliable for a short surface gap (apo equilibration relaxes a two- or three-residue loop anyway) and arbitrary for anything longer, where real loop modelling (Modeller, Rosetta) or a full predicted model (AlphaFold) is required. Disordered *termini* are not modelled at all: they bear no labels and would only add unrestrained flailing, so we cap them.

Method used: AlphaFold-loop grafting with restrained closure. Rather than let a fixer build loops blind, we graft the missing internal residues from an AlphaFold model and use the crystal for everything resolved. For each antigen we (i) sequence-align the crystal chain to its AlphaFold model to fix the numbering offset, (ii) measure the Cα-RMSD between model and crystal over all matched residues — the "is the prediction trustworthy here" test — and (iii) for each internal gap, locally superpose the model on the flanking resolved residues and splice in its loop at crystal numbering. `pdbfixer` then completes any partial side chains, and a short OpenMM minimization with all crystal atoms position-restrained (only the loop free, plus an explicit harmonic across the spliced junction) closes the peptide bonds. Disordered termini are capped, not modelled. Scripts: `analysis/repair_structure.py`, `finalize_structure.py`, `close_loops.py`; host-side env `env/repair-environment.yml` (Python 3.11, OpenMM 8.5, `pdbfixer`, run locally, not on Gemini).

Antigen (PDB, chain)	Unresolved segments (`REMARK 465`)	Repair outcome
hGH (`1HGU` A)	F1; **K38–E39** (internal); F191	AF-graft (AFDB `P01241`); **done**
Tissue factor (`1AHW` C)	S1–T3; **V83–G90** (8-res); Q212–E219	AF-graft (AFDB `P13726`); **done**
HCMV gB (`5C6T` A)	63–87 (25); 115–120; 219–220; 237–239; **437–468 (32)**	not in AFDB → AF3 prediction (pending)

For tissue factor the AlphaFold model agrees closely with the crystal (global Cα-RMSD 1.12 Å over 200 residues; the V83–G90 loop fits its flanks at 1.29 Å), so the 8-residue graft is well supported. For hGH only the 2-residue K38–E39 turn is grafted (local fit 1.81 Å); the larger global RMSD (3.36 Å, hinge flexibility of the four-helix bundle) is immaterial for so short a splice. Both repaired chains validate as continuous, with the expected disulfides (hGH 53–165, 182–189; TF 49–57, 186–209) and no steric clashes. HCMV gB is a viral protein *absent from AlphaFold DB*, and its largest gap

(32 residues) is true *de novo* prediction, not loop repair; it requires a fresh model. We generate one with **AlphaFold3** on Gemini (the lab's `alphafold_3.0.1` Singularity container and installed weights/databases; `md/5C6T/apo/af3.sbatch`, mirroring the episcaf pipeline), then graft its loops the same way. The AF3 model's agreement with the crystal over the resolved residues is the same trust check applied above; if it disagrees we simulate the AF3 ectodomain directly.

Consequences for the run queue. Eight of the nine non-pristine/pristine antigens are now ready to prep on Gemini (`git pull` then `sbatch`): pristine chains HEL (`1AKI`), HPr (`2JEL`), VEGF (`1BJ1`), IFN-γR (`1JRH`), Dengue E (`1OKE`), HA (`1HGG`); plus the repaired hGH (`1HGU`) and tissue factor (`1AHW`). Outstanding: HCMV gB (`5C6T`, awaiting an AF3 model) and the previously deferred MT-SP1 (§5, two internal residues, same AF-graft recipe) and Bont/A1.

10 Run log

Exact-command reproducibility ledger. Append a row per stage; pin the GROMACS version actually used on Gemini, seeds, and Slurm job id.

Date	Stage	Command	GROMACS ver.	Job / output
2026-06-18	structure	`curl .../1AKI.pdb`	—	`structures/1AKI.pdb`
pending	prep	`sbatch md/1AKI/apo/prep.sh`	2023.2-dev	out/npt.gro/cpt
pending	production	`sbatch md/1AKI/apo/md.sh`	2023.2-dev	out/md.xtc

Status (2026-06-18). Project scaffolded; benchmark audited; first system (HEL/1AKI) selected and label mapping verified 1:1; apo-HEL run #1 staged and signed off. Gemini environment confirmed: module `Gromacs` (capital G), GROMACS `2023.2-dev` (avx2_256), partition `gpu-a100` (4-day limit, A100 fastest); repo cloned to `/scratch/bneff/bcell_epitope`. Production walltime set to 48 h ceiling. **Next:** `git pull` on Gemini, then `sbatch prep.sh` → `sbatch md.sh`; record Slurm job ids and ns/day. Production length 100 ns (single trajectory). Open: replicas (deferred — random seeds make them trivial); scheduling the FRESEAN/3D-2PT run.

Run-script fixes (bringing up run #1 on Gemini). Three issues surfaced and were fixed while getting prep to run; recorded for reproducibility:

- **Module name.** Gemini's module is `Gromacs` (capital G); lowercase `gromacs` errors on a missing `shared` dependency. Scripts updated.
- **Config paths under Slurm.** Slurm copies the batch script to a private spool dir, so `$(dirname $0)/configs` resolved into the spool dir and `ions.mdp` was not found. Scripts now anchor paths to `$SLURM_SUBMIT_DIR`; **always submit from the repo root**.
- **GPU bonded offload on EM.** `-bonded gpu` requires a dynamical integrator; EM uses `steep`, so the EM `mdrun` now uses `-nb gpu` only. NVT/NPT/ production use `md` and keep `-nb/-pme/-bonded gpu` (audited).

Also removed the obsolete `ns_type` entry from all mdp files (ignored under the Verlet scheme). **Caveat:** Gemini scratch is not durable — move logs/outputs off scratch after each run (an earlier prep's logs were lost to a purge). Sanity check per run: `pdb2gmx.log` must report **4 disulfides** for HEL.

Expansion to multiple antigens (2026-06-21). Switched the box from dodecahedron to **cubic** (interpretability; uniform across systems) and re-ran HEL for consistency. Staged three further apo-antigen systems from the benchmark, hybrid-sourced (strip antigen chain(s) from the complex; numbering kept aligned to the labels). Provenance:

- **HPr** — `2JEL` chain P (res 1–85, no Cys); `md/2JEL/apo`.
- **VEGF** — `1BJ1` chains V+W (res 14–107, disulfide-linked homodimer; `pdb2gmx -merge all` for the inter-chain SS); `md/1BJ1/apo`.
- **Bont/A1 Hc** — `2NYY` chain A res 872–1295 (C-terminal receptor-binding domain only; catalytic-domain Zn/Ca excluded; res 872 is a domain cut → charged N-terminus); `md/2NYY/apo`.

All four (HEL+these) submitted overnight as prep→production chains (`sbatch --dependency=afterok`). **Deferred:** MT-SP1 (2 missing internal residues), Bont/A2 (homology models), hGH (provisional labels). Per-system disulfide expectations recorded in each `prep.sh`.

Structure repair (2026-06-29). Audited `REMARK 465` for the incomplete staged structures and repaired two of three by AlphaFold-loop grafting with restrained closure (method + RMSDs in §9; scripts `analysis/repair_structure.py`, `finalize_structure.py`, `close_loops.py`; host env `env/repair-environment.yml`). **hGH** (`1HGU`, K38–E39 grafted from AFDB `P01241`) and **tissue factor** (`1AHW`, loop V83–G90 from AFDB `P13726`; AF–crystal global Cα-RMSD 1.12 Å) are now frozen as `structures/{hGH_1HGU,TissueFactor_1AHW}_fixed.pdb` — continuous, correct disulfides, clash-free — so both are ready to prep. **HCMV gB** (`5C6T`) is *absent from AlphaFold DB* (viral) and its 32-residue gap is de-novo prediction; deferred pending an AF3 (AlphaFold Server) model. Net: eight antigens ready to `sbatch` on Gemini.

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